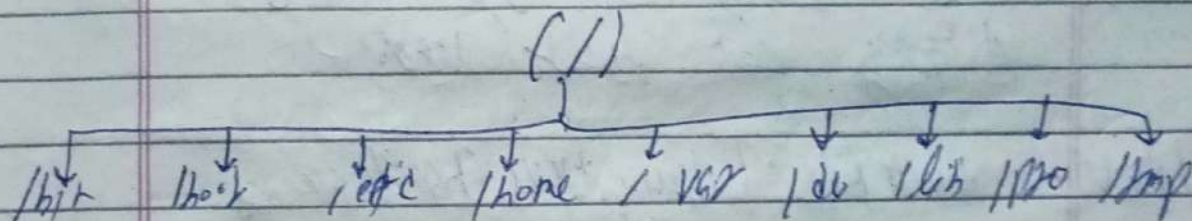


## DAY-07 Linux File System Hierarchy & Scenario-Based Practice

### PART 1: Linux File System Hierarchy

The Linux file system hierarchy is a standard directory structure that organizes files and data on storage devices, starting from the root directory (/):



\* / → The root directory, the top of the hierarchy. Everything starts from this.

\* /bin → Essential binary program (like ls, cp)  
[mv, cat, etc.]

\* /sbin → System Administration Commands  
[used for managing and repairing the system.]

Example:-

/sbin/xboot  
/sbin/fsck  
/sbin/shutdown

\* /boot → This directory we use when your system is startup:-



\* /etc - This is for configuration files.  
 Exmpl:-

/etc/passwd  
 /etc/hosts

\* /home - User home directories -  
 → The stores personal files  
 Exmpl:- /home/sunny

\* /root - This directory is for Home  
 directory of root user:-

\* /dev - device files: [link to hardware of  
 files]

Exmpl:-  
 /dev/sda  
 /dev/hdd

\* /proc - virtual file system:-  
 contain process and kernel information

\* you find here primary memory  
 " " " ioperm, ioperm;  
 all the process related files you are  
 here find!

Exmpl:-  
 ls -l /proc/meminfo

\* /sys  
→ It contain kernel and hardware information.

Interface between kernel and device.  
Example:-

ls /sys  
Contains:  
block  
bus  
class  
dev  
kernel

\* /tmp (Temporary)  
This directory contain temporary files.  
Example:

cd /tmp  
ls -l demo.txt

\* /var → variable  
The variable data files such as log files are located in the /var directory.

- /var/log: System log files generated by OS and other applications.
- /var/lib: Contains database and package
- /var/mail: Contains Emails
- /var/tmp: Contains temporary files



\* /usr :- The usr directory contains  
(user binaries) applications and files used by  
user, as opposed to application  
and app files used by the  
system.

↳ /usr/bin :- shared binaries  
↳ /usr/bin :- user binaries [hard disk]  
:- Command used by all the  
user of the system  
located here. [python, etc]

↳ /sbin :- System binaries :-  
Command used by the system  
administrator, for system  
maintenance purpose.  
Example:- ifconfig, reboot, fdisk,  
etc.

\* /opt :- optional third-party software  
Example:- /opt/google  
/opt/jre

\* /mnt :- Temporary mount point.  
This directory is used to mount a  
file system temporarily.

\* /media :- Removable media devices

The media directory contains subdirectories  
where removable media devices inserted  
into the computer are mounted.



\* 24 -s /var/log/\* 27 dev/hull /sort -h /tail -s

\* 24 -s /var/log/\*

12 show to ~~list~~ all files that can use disk space

(17) 27 dev/hull [error all time]

map:-

0 key keyboard input  
1 normal output  
2 error output

⇒ /dev/hull  
special device that discards everything  
anything sent there disappears -  
echo "hello" > /dev/hull

(17) 1 sort -h

pipe operator. Pass out from one command to another.

sort :- sort lines

-h :- human-readable sort

10k

50M

25%

without -h

→ 10k

248

50M

(17) tail : display last lines

-s : show last 5 lines.

[tail -h 5]

\* cat /etc/passwd  
when the hostname used to  
identify the system on a network

\* ls -la ~

This command all files and  
directories with hidden files in your home  
directory

$\{ a = /home/alan \}$

~ = This is shortcut



Page No. \_\_\_\_\_  
Date: \_\_\_\_\_

## Part 2 - Scenario - Based Practice

Example Scenario: - Check if a service is running.

Step 1: - Check service status

System status type: -

This command is to check if the type service is running or not.

Step 2: - systemctl list-units --type=services  
List all service whether it is running or failed

Step 3: - Check if service is enabled or on the system or is enabled type

[Scenario: - Service not starting]

Step 1: - systemctl status nginx

This command is to find running or not

Step 2: - systemctl is-enabled nginx

This command is to show if it is enabled or not. To know if it will start automatically after reboot.

Step 4: - journalctl -u nginx

Journalctl -u nginx -h 50

This command shows the log of the type and how last 50 lines

[Scenario 2:- High CPU usage]

Step 1:- top :-

This command is used to check real time CPU and Mem usage

Step 2: ps aux --sort=-%cpu | head -5  
This command is used to show available in the CPU usage high to low (-) top 5 process.

Key point (-) :- use for high to low  
(+) :- use for low to high

Step 3:- ps -o pid,ppid,pname,pcpu,cmsize -p pids

This command is show a particular process information about ppid, pcpu, mem, CPU, for each pid.

ps aux --sort=-%cpu | head -1 # Highest CPU

ps aux --sort=-%cpu | head -1 # Highest CPU

ps aux --sort=-%mem | head -1 # Highest memory

ps aux --sort=-%mem | head -1 # Lowest memory



\* Scenario 3: Finding service logs:-

Step 1:- systemctl status ssh

Check if the service is running or not.

Step 2:- journal -4 ssh -4 50

This command is used to list 50 lines of log ssh.

Step 3: journalctl -4 ssh -f

To see real-time log of ssh:-

PAGE NO. \_\_\_\_\_  
DATE: \_\_\_\_\_

## [ Scenario 4: File Permission Issue ]

Step 1: check

This script - backup.sh is not  
execute.

Step 1: ls -l backup.sh

This step command the use to  
show the permission of the file

Step 2: add execute permission  
at chmod +x backup.sh

This command execute permission  
for all the users.

Step 3: verify ls -l backup.sh

Step 4: Try to run it

Command: ./backup.sh